Adaptive Forget Gates in Long Short-Term Memory Networks (Gers, Schmidhuber & Cummins 2000)

Adding a multiplicative forget gate to the LSTM constant error carousel enables recurrent networks to reset internal state dynamically, solving continual time-series tasks where standard LSTM fails.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL, ORGANIZATIONAL, PHILOSOPHICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Standard Long Short-Term Memory (LSTM) networks with fixed unit self-connections accumulate internal cell state indefinitely on continual input streams without explicit sequence boundaries; introducing a learned multiplicative forget gate ($f \in (0, 1)$) allows the cell to flush obsolete state and prevents saturation.

EVIDENCE AND CAUSAL ASSESSMENT

Hochreiter and Schmidhuber (1997) solved vanishing gradients via the Constant Error Carousel (CEC), which sets the recurrent cell self-weight to $1.0. Gers et al. (2000) showed that on continual streams lacking external reset markers, the CEC state grows unboundedly, saturating the output squashing function and collapsing gradients. Replacing the fixed $1.0 connection with a learned sigmoidal forget gate restored continual learning.

EFFECT SIZE AND PERSISTENCE

On continual embedded Reber grammar streams without reset markers, standard 1997 LSTM converged in 0\% of trials as cell states saturated, whereas forget-gate LSTM converged reliably (>90\% of trials) across thousands of continuous symbols.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as robust (<20%) across controlled architectural ablations. Performance still depends on forget-gate bias initialization (initializing forget biases to +1.0 avoids early gradient attenuation before the gate learns what to retain).

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

A neural forget gate does not perform semantic or psychological repression, nor does it guarantee privacy-compliant data erasure (machine unlearning), because learned information remains distributed across shared network weights even after a transient cell activation c is zeroed.

WHAT ACTION IS REASONABLY JUSTIFIED

Stateful sequence architectures operating over continuous streams without external episode resets must incorporate explicit, bounded state-decay or forget-gating mechanisms, with positive initial retention bias during early training.

VERIFIED SOURCES & DOI IDENTIFIERS

- Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation, 9(8), 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735

- Gers, F. A., Schmidhuber, J., & Cummins, F. (2000). Learning to Forget: Continual Prediction with LSTM. Neural Computation, 12(10), 2451–2471. https://doi.org/10.1162/089976600300015015